~~P9 1.1~~

P9 1.5

(a)

$$A^2 \int_{-\infty}^{+\infty} e^{-2\lambda|x|}\,dx = 1 \qquad \int_{-\infty}^{+\infty} e^{-2\lambda|x|}\,dx = 2\int_0^{\infty} e^{-2\lambda x}\,dx = \frac{2}{-2\lambda}(0-1) = \frac{1}{\lambda}$$

$$A^2/\lambda = 1, \quad A = \sqrt{\lambda}$$

$$\psi(x,t) = \sqrt{\lambda}\, A e^{-\lambda|x|} e^{-iwt}$$

(b)

$$\langle x\rangle = \int_{-\infty}^{+\infty} \psi^* x \psi(x)\,dx = A^2 \int_{-\infty}^{+\infty} x e^{-2\lambda|x|}\,dx = 0$$

$$\langle x^2\rangle = \int_{-\infty}^{+\infty} \psi^* x^2 \psi\,dx = A^2 \int_{-\infty}^{+\infty} x^2 e^{-2\lambda|x|}\,dx$$

$$= 2A^2 \int_0^{\infty} x^2 e^{-2\lambda x}\,dx = \left(x^2 + \frac{1}{\lambda}x + \frac{1}{2\lambda^2}\right) e^{-2\lambda x} \cdot \frac{1}{-2\lambda}\Big|_0^{\infty}$$

x^3 $3x^2$ $6x^1$

+	x^2	$e^{-2\lambda x}$	
−	$2x$	$\frac{1}{-2\lambda} e^{-2\lambda x}$	
$\frac{n!}{(n-k)!}$	2	$\frac{1}{4\lambda^2} e^{-2\lambda x}$	
−	0	$\frac{1}{-8\lambda^2} e^{-2\lambda x}$	

$$= \left(\frac{x^2}{-2\lambda} - \frac{2x}{4\lambda^2} - \frac{1}{4\lambda^3}\right) e^{-2\lambda x} \cdot 2\lambda\Big|_0^{\infty}$$

$$= \frac{1}{4\lambda^3} \cdot 2\lambda = \frac{1}{2\lambda^2}$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2}$$

$$f^1(x) = (x^n)' \qquad f^k(x) = \frac{n!}{(n-k)!} x^{n-k}$$

$$\int_0^{\infty} x^n e^{-ax}\,dx = \left[\sum_{k=1}^{n} \frac{n!}{(n-k)!} x^{n-k} \frac{1}{(-a)^{k+1}}\right] e^{-ax}\Big|_0^{\infty}$$

$$\frac{n!}{a^{n+1}} = 0 - n!\,\frac{1}{(-a)} = \left[\sum_{k=0}^{n} \frac{n!}{(n-k)!} x^{(n-k)} \frac{1}{(-a)^{k+1}}\right] e^{-ax}\Big|_0^{\infty}$$

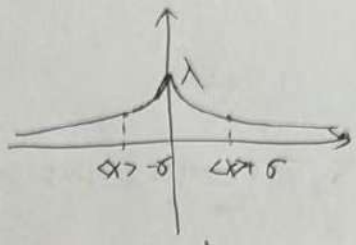
$$= (-1)^{k+1} n!\,(-a)^{-k-1} = \frac{n!}{a^{k+1}}$$

P 9 1.5

(c)

$$\sigma = \sqrt{\langle x^2\rangle - \langle x\rangle^2} = \frac{1}{\sqrt{2}\lambda}$$

$$|\psi|^2 = \lambda e^{-2\lambda|x|}$$



$$\langle x\rangle + \sigma = \frac{1}{\sqrt{2}\lambda}$$

$$\langle x\rangle - \sigma = -\frac{1}{\sqrt{2}\lambda}$$

$$P_{in} = \int_{-\frac{1}{\sqrt{2}\lambda}}^{\frac{1}{\sqrt{2}\lambda}} \psi^* e^{-2\lambda|x|}\, dx = 2\int_0^{\frac{1}{\sqrt{2}\lambda}} \lambda e^{-2\lambda x}\, dx$$

$$= 2\lambda \frac{1}{-2\lambda} e^{-2\lambda x}\Big|_0^{\frac{1}{\sqrt{2}\lambda}} = -(e^{-\sqrt{2}} - 1) = 1 - e^{-\sqrt{2}}$$

$$P_{out} = e^{-\sqrt{2}}$$

P 12 1.7

$$\langle p\rangle = \int_{-\infty}^{+\infty} \psi^* \left(-i\hbar\frac{\partial}{\partial x}\right)\psi\, dx = -i\hbar\int_{-\infty}^{+\infty} \psi^* \frac{\partial}{\partial x}\psi\, dx$$

$$\frac{d\langle p\rangle}{dt} = -i\hbar\int_{-\infty}^{+\infty}\left(\frac{\partial\psi^*}{\partial t}\frac{\partial\psi}{\partial x} + \psi^*\frac{\partial^2\psi}{\partial t\partial x}\right)dx$$

$$i\hbar\frac{\partial\psi}{\partial t} = \left[-\frac{\hbar^2\partial^2}{2m\partial x^2} + V\right]\psi \Rightarrow \frac{\partial\psi}{\partial t} = \left[\frac{i\hbar\partial^2}{2m\partial x^2} - \frac{i}{\hbar}V\right]\psi$$

$$\Rightarrow \frac{\partial\psi^*}{\partial t} = \left(\frac{-i\hbar}{2m}\frac{\partial^2}{\partial x^2} + \frac{i}{\hbar}V\right)\psi^*$$

$$\left(\frac{-i\hbar}{2m}\frac{\partial^2}{\partial x^2} + \frac{i}{\hbar}V\right)\psi^*\frac{\partial\psi}{\partial x} + \psi^*\frac{\partial}{\partial x}\left[\frac{i\hbar}{2m}\frac{\partial^2}{\partial x^2} - \frac{i}{\hbar}V\right)\psi\Big]$$

$$= \frac{-i\hbar}{2m}\frac{\partial^2}{\partial x^2}\psi^*\frac{\partial\psi}{\partial x} + \frac{i}{\hbar}V\psi^*\frac{\partial\psi}{\partial x} + \psi^*\frac{i\hbar}{2m}\frac{\partial^3}{\partial x^3}\psi - \psi^*\frac{i}{\hbar}V\frac{\partial\psi}{\partial x}$$

$$\frac{d\langle p\rangle}{dt} = \int \left[\frac{-\hbar^2}{2m}\frac{\partial^2\psi^*}{\partial x^2}\frac{\partial\psi}{\partial x} + V\psi^*\frac{\partial\psi}{\partial x} + \psi^*\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} - \psi^*\frac{\partial(V\psi)}{\partial x}\right] dx$$

$$= \int \frac{\partial^2\psi^*}{\partial x^2}\frac{\partial\psi}{\partial x}\,dx = \left[\frac{\partial\psi^*}{\partial x}\frac{\partial\psi}{\partial x}\right]_{-\infty}^{+\infty} - \int \frac{\partial\psi^*}{\partial x}\frac{\partial^2\psi}{\partial x^2}\,dx$$

$$= -\int_{-\infty}^{+\infty} \frac{\partial\psi^*}{\partial x}\frac{\partial^2\psi}{\partial x^2}\,dx$$

$$\int_{-\infty}^{+\infty} \psi^*\frac{\partial^3\psi}{\partial x^3}\,dx = \left[\psi^*\frac{\partial^2\psi}{\partial x^2}\right]_{-\infty}^{+\infty} - \int_{-\infty}^{+\infty}\frac{\partial\psi^*}{\partial x}\frac{\partial^2\psi}{\partial x^2}\,dx$$

$$= -\int_{-\infty}^{+\infty} \frac{\partial\psi^*}{\partial x}\frac{\partial^2\psi}{\partial x^2}\,dx$$

$$\frac{d\langle p\rangle}{dt} = \int_{-\infty}^{+\infty}\left[\frac{\partial(V\psi)}{\partial x}^*\psi - \frac{\partial\psi}{\partial x}^*V\psi\right]dx$$

$$= \int_{-\infty}^{+\infty} -\psi^*\psi\frac{\partial V}{\partial x}\,dx = -\left\langle\frac{\partial V}{\partial x}\right\rangle$$

P13 1.9

(a) $\int_{-\infty}^{+\infty}\psi^*\psi\,dx = A^2\int_{-\infty}^{+\infty} e^{-2am x^2/\hbar}\,dx = A^2\sqrt{\frac{\pi\hbar}{2ma}} = 1 \quad A = \sqrt{\frac{2m}{\pi\hbar}}$

(b) $i\hbar\frac{\partial\psi}{\partial t} = \left(\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V\right)\psi$

$$\frac{\partial\psi}{\partial x} = -2max\psi$$

$$\frac{\partial}{\partial x}\left(\frac{\partial\psi}{\partial x}\right) = -\frac{2ma}{\hbar}\psi + \frac{\psi m^2a^2x^2\psi}{\hbar^2}$$

$$\hbar a\psi = -\frac{\hbar^2}{2m}\left(\frac{-2ma}{\hbar}\psi + \frac{\psi m^2a^2x^2}{\hbar^2} + V\right)\psi$$

即 $\hbar a = \hbar a - 2ma^2x^2 + V(x) \qquad V(x) = 2ma^2x^2$

(c) 易知 $\langle x\rangle = 0 \quad \langle p\rangle = 0$

$$\langle x^2\rangle = \int_{-\infty}^{+\infty} x^2|\psi|^2\,dx = A^2\int_{-\infty}^{+\infty} x^2 e^{-\frac{2am}{\hbar}x^2}\,dx$$

$$= \left(\frac{2ma}{\pi\hbar}\right)^{\frac{1}{2}}\frac{\hbar}{4am}\sqrt{\frac{\pi\hbar}{2am}} = \frac{\hbar}{4am}$$

$$= \frac{i\hbar}{2m}\int_a^b \left(\psi^*\frac{\partial^2\psi}{\partial x^2}\right.$$

$$\frac{d\langle p\rangle}{dt} = \int \frac{-\hbar^2}{2m} \frac{\partial^3 \psi^*}{\partial x^3} \frac{\partial \psi}{\partial x} + V\psi^* \frac{\partial \psi}{\partial x}$$

$$\langle p^2\rangle = \int_{-\infty}^{+\infty} \psi^* (-\hbar^2 \frac{\partial^2}{\partial x^2}) \psi \, dx$$

$$= -\hbar^2 \int_{-\infty}^{+\infty} \psi^* \frac{\partial^2 \psi}{\partial x^2} dx = -\hbar \int_{-\infty}^{+\infty} \left(-2\frac{ma}{\hbar} x \right) \psi + \frac{4m^2a^2x^2}{\hbar^2} \psi \psi^* dx$$

$$= \int_{-\infty}^{+\infty} (2ma\hbar - 4m^2a^2x^2) |\psi|^2 dx$$

$$= \int 2ma\hbar |\psi|^2 dx - \int_{-\infty}^{+\infty} 4m^2a^2x^2 |\psi|^2 dx$$

$$= 2ma\hbar - 4m^2a^2 \cdot \frac{\hbar}{4am} = 2ma\hbar - ma\hbar = ma\hbar$$

(d) $\sigma_x = \sqrt{\langle x^2\rangle - \langle x\rangle^2} = 0 \sqrt{\frac{\hbar}{4am}}$

$\sigma_p = \sqrt{\langle p^2\rangle - \langle p\rangle^2} = \sqrt{am\hbar}$

$\sigma_x \sigma_p = \frac{\hbar}{2}$ 满足

P14 1-14

$$\frac{dP_{ab}}{dt} = \int_a^b \frac{\partial}{\partial t} (\psi^* \psi) dx = \int_a^b \left(\psi^* \frac{\partial \psi}{\partial t} + \psi \frac{\partial \psi^*}{\partial t} \right) dx$$

$$i\hbar \frac{\partial \psi}{\partial t} = \left(-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V \right) \psi \Rightarrow \frac{\partial \psi}{\partial t} = \frac{i\hbar}{2m} \frac{\partial^2 \psi}{\partial x^2} - \frac{i}{\hbar} V\psi$$

$$\frac{\partial \psi^*}{\partial t} = -\frac{i\hbar}{2m} \frac{\partial^2 \psi^*}{\partial x^2} + \frac{i}{\hbar} V\psi^*$$

代入 $\frac{dP_{ab}}{dt} = \int_a^b \left(\frac{i\hbar}{2m} \psi^* \frac{\partial^2 \psi}{\partial x^2} - \frac{i}{\hbar} \psi^* V\psi - \frac{i\hbar}{2m} \psi \frac{\partial^2 \psi^*}{\partial x^2} + \frac{i}{\hbar} \psi V\psi^* \right) dx$

$$= \frac{i\hbar}{2m} \int_a^b \left(\psi^* \frac{\partial^2 \psi}{\partial x^2} - \psi \frac{\partial^2 \psi^*}{\partial x^2} \right) dx$$

$$= \frac{i\hbar}{2m} \int_a^b \left[\frac{\partial}{\partial x} \left(\psi^* \frac{\partial \psi}{\partial x} - \psi \frac{\partial \psi^*}{\partial x} \right) \right] dx = \frac{i\hbar}{2m} \left[\psi^* \frac{\partial \psi}{\partial x} - \psi \frac{\partial \psi^*}{\partial x} \right]_a^b$$

$J(x,t) = \frac{i\hbar}{2m}(\psi \frac{\partial \psi^*}{\partial x} - \psi^* \frac{\partial \psi}{\partial x})$，故

$\frac{dP_{ab}}{dt} = J(a,t) - J(b,t)$

$\int_{-\infty}^{+\infty} |\psi|^2 dx = 1$，$[\psi] = L^{-\frac{1}{2}}$，$[\frac{\hbar}{m}] = L^2 T^{-1}$

$[\frac{\hbar}{m}] = \frac{J \cdot S}{kg} = \frac{kg \cdot m^2 \cdot s^{-2} \cdot S}{kg} = m^2 s^{-1} = L^2 T^{-1}$

$[\psi \frac{\partial \psi^*}{\partial x}] = L^{-\frac{1}{2}} \cdot L^{-\frac{3}{2}} = L^{-2}$ 故 $[J] = T^{-1} = S^{-1}$

(b) $A e^{-\alpha[mx^2/\hbar) + it]} = \psi(x,t)$ $A = (\frac{2m\alpha}{\pi\hbar})^{\frac{1}{4}}$

$J(x,t) = \frac{i\hbar}{2m}(\psi(-\frac{2m\alpha x}{\hbar})\psi^* - \psi^*(-\frac{2m\alpha x}{\hbar}\psi))$

$= \frac{i\hbar}{2m} \frac{2m\alpha x}{\hbar} \psi^* \cdot 0 = 0$

P. 15 1.17.

(a) $\int_{-\infty}^{+\infty} \psi^* \psi \, dx = \int_{-a}^{a} A^2(a^2 - x^2)^2 = 2\int_0^a (a^2 - x^2)^2 dx = 2A^2[a^2 x - \frac{1}{3}x^3]\Big|_0^a$

$= 2A \cdot (a^3 - \frac{1}{3}a^3) = \frac{4}{3}a^3 A = 1 \quad a = (\frac{3}{4})^{\frac{1}{3}}$

$= 2A^2 \int_0^a (a^4 - 2a^2x^2 + x^4) dx$

$= 2A^2 [a^4 x - \frac{2a^2}{3}x^3 + \frac{1}{5}x^5]_0^a = 2A^2(a^5 - \frac{2a^5}{3} + \frac{1}{5}a^5)$

$= 2A^2 \cdot \frac{8}{15}a^5 = 1 \quad A^2 = \frac{15}{16a^5} \quad A = \frac{\sqrt{15}}{4a^{\frac{5}{2}}} = \frac{1}{4a^2}\sqrt{\frac{15}{a}}$

(b) $\langle x \rangle = \int_{-\infty}^{+\infty} \psi^* x \psi \, dx = A^2 \int_{-\infty}^{+\infty} x(a^2 - x^2)^2 dx = 0$

(c) $\hat{p} = -i\hbar \frac{d}{dx}$

$\frac{i\hbar}{2m} \int_a^b [\frac{\partial}{\partial x}(\psi^* \frac{\partial \psi}{\partial x} - \psi \frac{\partial \psi^*}{\partial x})] dx = \frac{i\hbar}{2m}[\psi^* \frac{\partial \psi}{\partial x} - \psi \frac{\partial \psi^*}{\partial x}]_a^b$

$$\langle p\rangle = \int_{-\infty}^{+\infty} \psi^* \left(-i\hbar \frac{d}{dx}\right) \psi \, dx$$

$$= -i\hbar \int_{-\infty}^{+\infty} A(a^2-x^2)\cdot(-2Ax)\,dx = 2i\hbar A^2 \int_{-a}^{a} x(a^2-x^2)\,dx = 0$$

$\langle p\rangle = m\frac{d\langle x\rangle}{dt}$，因为 $\langle x\rangle$ 为与 t 无关的常数，无法求导。

(d)

$$\langle x^2\rangle = A^2\int_{-a}^{a} x^2(a^2-x^2)^2\,dx = A^2\int_{-a}^{a}(a^4x^2 - 2a^2x^4 + x^6)\,dx$$

$$= 2A^2\left[\frac{a^4}{3}x^3 - \frac{2a^2}{5}x^5 + \frac{1}{7}x^7\right]_0^a = \frac{16}{105}A^7a^7 = \frac{a^2}{7}$$

(e)

$$\langle p^2\rangle = \int_{-a}^{a} \psi^*\left(-\hbar^2\frac{d^2}{dx^2}\right)\psi\,dx = \int_{-a}^{a} -2A\hbar^2(a^2-x^2)\,dx$$

$$\frac{d^2\psi}{dx^2} = \frac{d}{dx}(-2Ax) = -2A,$$

$$= 2A^2\hbar^2\int_{-a}^{a}(a^2-x^2)\,dx = \frac{8\hbar^2A^2a^3}{3} = \frac{5\hbar^2}{2a^2}$$

(f) (g) (h)

$$\sigma_x = \sqrt{\langle x^2\rangle - \langle x\rangle^2} = \frac{a}{\sqrt{7}}$$

$$\sigma_p = \sqrt{\langle p^2\rangle - \langle p\rangle^2} = \sqrt{\frac{5\hbar^2}{2a^2} - 0} = \sqrt{\frac{5}{2}}\frac{\hbar}{a}$$

$$\sigma_x\sigma_p = \sqrt{\frac{5}{14}}\hbar \geq \frac{1}{2}\hbar$$ 不符合